COLLECTION OF THE CTL DECISIONS
DECISION SHEET

Standard: IEC 60950-1:2000, 1st ed,	**Sub clause:** 7	**Sheet No.** 445
Subject: Protection of users	**Key words:** Antenna, voltage surge test	**Decision taken at the 40th meeting 2003**

Question:

For protection of users and service persons from transients and voltage surges, what are the appropriate requirements between the following points for ITE connected to an external antenna:

(a) antenna connection and Primary circuits?
(b) antenna connection and SELV circuits when the antenna cabling is coax?
(c) Antenna connection and SELV circuits when the antenna cabling is unshielded twisted pair (UTP)?

See attached diagram.

Decision:

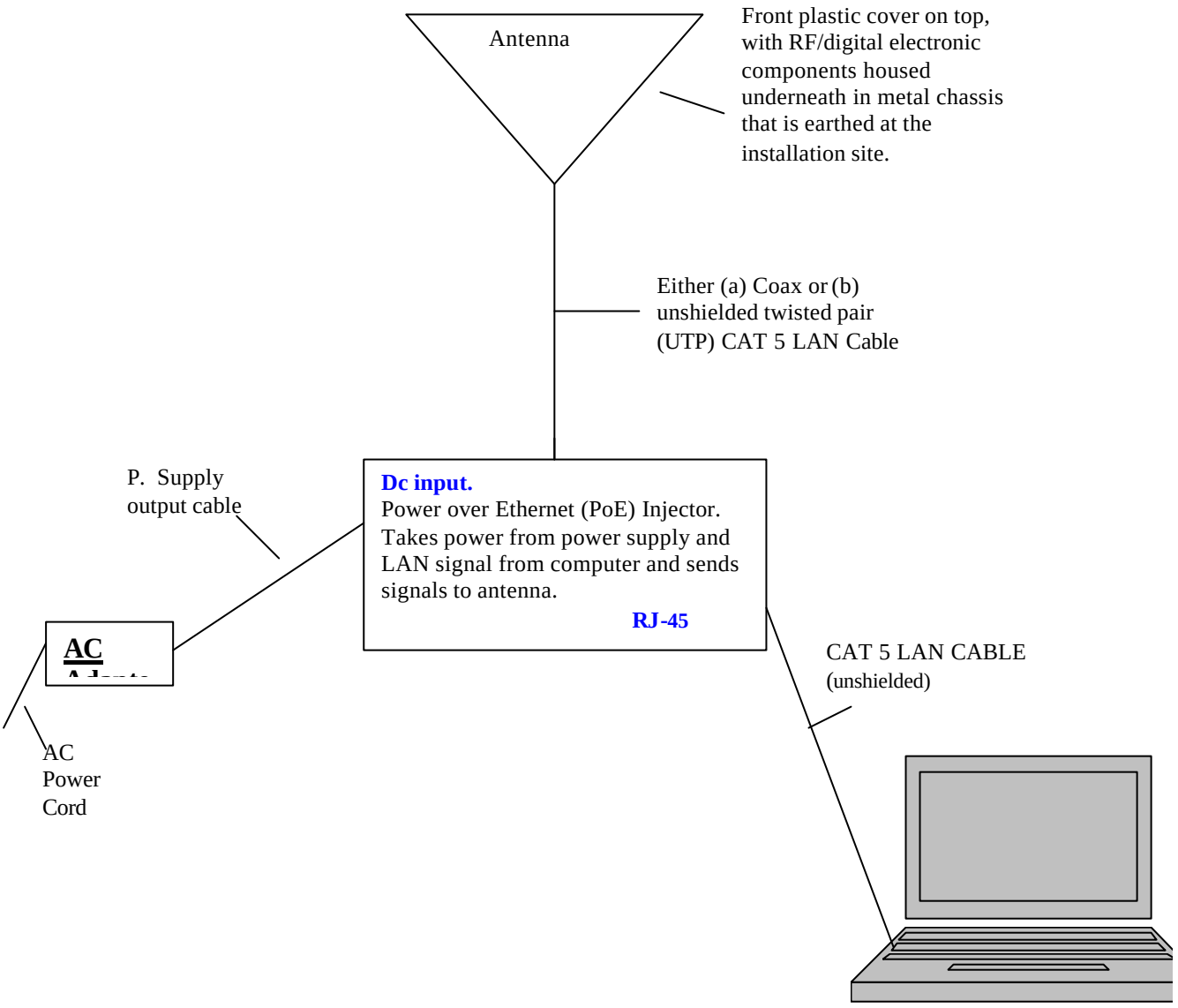
Since Clause 7 is in IEC 60950-1, but is not in IEC 60950 3rd Ed, all references are to IEC 60950-1

The circuit connected to the antenna is classified in accordance with Sub-Clause 7.1 and subjected to the appropriate requirements per 7.1. Additionally,

(a) Sub-clause 7.3.1 is applied between the external antenna connection (terminal) and Primary circuits.
(b) Sub-clause 7.2 is applied between the external antenna connection (terminal) and the SELV connector (RJ-45) when the antenna cabling is coax.
(c) Sub-clause 7.2 is applied between the external antenna connection (terminal) and the SELV connector (RJ-45) when the antenna cabling is UTP.

Explanatory notes:

For the same question related to IEC 60950 3rd Ed. ref. to the relevant decision sheet for that standard is made. Based on the answers received on the enquiry and a discussion with the chairman of IEC TC108, the decision for IEC 60950 3rd Ed. has to be different.


Antenna
Front plastic cover on top, with RF/digital electronic components housed underneath in metal chassis that is earthed at the installation site.
Either (a) Coax or (b) unshielded twisted pair (UTP) CAT 5 LAN Cable
P. Supply output cable
Dc input.
Power over Ethernet (PoE) Injector. Takes power from power supply and LAN signal from computer and sends signals to antenna.
RJ-45
AC
AC Power Cord
CAT 5 LAN CABLE (unshielded)